

Alex (Wei) Chen

Director

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Summary

Director at Qualcomm (via acquisition of Nexa AI). Researcher and engineer focused on AI systems, LLM/VLM inference optimization, hardware-software co-design, and robotics. PhD from Stanford University (2024).

Experience

Qualcomm , Director (Principal Engineer/Manager)	Santa Clara, CA
- Joined Qualcomm following the acquisition of Nexa AI. Lead on-device AI optimization research and engineering focused on accelerating generative AI inference on Qualcomm Snapdragon NPU (Hexagon HTP) and edge SoCs. - Drive end-to-end optimization of LLMs and VLMs for Qualcomm Hexagon HTP NPU, encompassing hardware-aware quantization (INT4/INT8/FP16 mixed-precision), operator fusion, memory bandwidth scheduling, and kernel-level tuning. - Collaborate with silicon, compiler, and runtime teams to co-design neural network architectures and inference pipelines that fully exploit Hexagon NPU vector and tensor acceleration. - Manage a cross-functional team spanning model research, runtime optimization, and SDK development.	2026 – present 1 year
Nexa AI , Founder, CEO and Chief Scientist	Palo Alto, CA
- Founded Nexa AI, specializing in efficient AI research and deployment. Built the NexaML engine and NexaSDK (7.6K GitHub stars) for generative AI model deployment on NPU, GPU, and CPU. Acquired by Qualcomm in 2026. - Principal architect and first author of AutoNeural, Octopus V1–V4, OmniVLM, OmniAudio, and NexaQuant. Octopus V2 represents ~2% of total HuggingFace downloads since 2022. - Solutions deliver ~55% faster throughput and ~28% better output quality vs. existing AI inference stacks. Trusted by Geely, HP, Lenovo, and İşbank. - Official partner of Qualcomm, AMD, NVIDIA, IBM, Google, Microsoft, Intel, Docker, HP, Lenovo, and Dell.	2024 – 2026 2 years

Education

Stanford University	California, US
Research: Large-scale numerical simulation frameworks combining HPC with ML-based surrogate models to accelerate PDE solvers.	2019 – 2024

Awards

Best Paper Runner-Up Award	Jan 2025
For the paper: DP-FedLoRA: Privacy-Enhanced Federated Fine-Tuning for On-Device Large Language Models. IEEE International Conference on Data Mining (ICDM 2025)	

Publications

Octopus: On-device language model for function calling of software APIs
Wei Chen, Zhiyuan Li, Mingyuan Ma
aclanthology.org/2025.naacl-industry.27

Octo-planner: On-device Language Model for Planner-Action Agents

Wei Chen, Zhiyuan Li, Zhen Guo, Yikang Shen

arxiv.org/abs/2406.18082

DP-FedLoRA: Privacy-Enhanced Federated Fine-Tuning for On-Device LLMs

Honghui Xu, Shiva Shrestha, Wei Chen, Zhiyuan Li, Zhipeng Cai

arxiv.org/abs/2509.09097

AutoNeural: Co-Designing Vision-Language Models for NPU Inference

Wei Chen, Liangmin Wu, Yunhai Hu, Zhiyuan Li, et al.

arxiv.org/abs/2512.02924

OmniVLM: A Token-Compressed, Sub-Billion-Parameter Vision-Language Model for Efficient On-Device Inference

Wei Chen, Zhiyuan Li, Shuo Xin

arxiv.org/abs/2412.11475

Dolphin: Long Context as a New Modality for Energy-Efficient On-Device Language Models

Wei Chen, Zhiyuan Li, Shuo Xin, Yihao Wang

arxiv.org/abs/2408.15518

Octopus v4: Graph of language models

Wei Chen, Zhiyuan Li

arxiv.org/abs/2404.19296

Octopus v3: Technical Report for On-device Sub-billion Multimodal AI Agent

Wei Chen, Zhiyuan Li

arxiv.org/abs/2404.11459

Octopus v2: On-device language model for super agent

Wei Chen, Zhiyuan Li

arxiv.org/abs/2404.01744

On-Device Language Models: A Comprehensive Review

Jiajun Xu, Zhiyuan Li, Wei Chen, Qun Wang, Xin Gao, Qi Cai, Ziyuan Ling

arxiv.org/abs/2409.00088

Skills

Programming Languages

Machine Learning

Hardware Acceleration

Systems

Languages

English

Fluent

Chinese (Mandarin)

Native